

# Supplementary Materials: Validating Experiments in Human-Robot Interaction: A State-of-the-Art Assessment

Anonymous Author(s)

## ACM Reference Format:

Anonymous Author(s). 2027. Supplementary Materials: Validating Experiments in Human-Robot Interaction: A State-of-the-Art Assessment. In *Proceedings of the 22nd Annual ACM/IEEE International Conference on Human-Robot Interaction (HRI '27)*. ACM, New York, NY, USA, 7 pages. <https://doi.org/XXXXXXX.XXXXXXX>

This supplementary document contains three sections. First, Section 1 provides additional detail on the seven dimensions of the validation framework. Second, Section 2 provides the complete codebook, including variable definitions and operational coding criteria used in data extraction. Third, Section 3 provides additional information on papers whose study settings or robot interaction modalities were coded as involving multiple categories or as not reported and were therefore excluded from the corresponding percentage calculations in the main manuscript.

The repo also contains the coding dataset and analysis code used to generate the reported results.

- **Coding dataset (.csv):** Contains the coded methodological indicators for the 73 papers.
- **Analysis script (.Rmd):** Contains the R code used to process the coded dataset and generate the descriptive statistics.

## 1 HRI Experiment Validation Framework

The HRI Experiment Validation Framework comprises seven dimensions: theoretical grounding, internal validity, instrumentation validity, manipulation validity, external validity, statistical conclusion validity, and research transparency.

### 1.1 Theoretical Grounding

Theoretical grounding concerns whether relationships examined in an experiment are motivated by a priori theoretical or logical reasoning. Statistical relationships alone establish neither why an experimental manipulation should affect an outcome nor the conditions under which a relationship should be expected to occur. Explicit theoretical reasoning therefore helps connect individual experimental findings to a broader cumulative understanding of human-robot interaction.

Theoretical grounding does not require every experiment to invoke an established or named “Big T” theory. Dennis and Valacich [10] distinguish such theories from more focused “little t” theoretical explanations consisting of clearly specified constructs, proposed

relationships among those constructs, and logical explanations for why those relationships should occur. We therefore adopt a broad definition of theoretical grounding that recognizes either established theoretical frameworks or explicit logical arguments supporting proposed relationships.

A priori specification is also relevant because explanations developed after observing results can be difficult to distinguish from predictions made before empirical testing, a practice associated with Hypothesizing After the Results are Known (HARKing) [15]. Accordingly, the framework examines whether explicit hypotheses are stated, whether hypothesized relationships are supported by theoretical or logical reasoning, and whether hypotheses appear before the Results section.

### 1.2 Internal Validity

Internal validity concerns whether observed differences in an outcome can reasonably be attributed to the experimental manipulation rather than to plausible rival explanations [5, 21]. Experimental designs strengthen internal validity by minimizing systematic differences between conditions other than the manipulated variable.

Random assignment is a primary mechanism for reducing systematic pre-existing differences between experimental groups in between-subject designs. In within-subject designs, the repeated exposure of participants to multiple experimental conditions introduces potential order and carryover effects. Randomization or counterbalancing of condition order can reduce systematic confounding associated with these effects [10, 14].

Our assessment therefore focuses on whether between-subject experiments report random assignment to experimental conditions and whether within-subject experiments report randomization or counterbalancing of condition order. Although procedural standardization is also important for internal validity, it was not coded because the level of procedural detail provided in published papers did not permit this practice to be assessed consistently.

### 1.3 Instrumentation Validity

Instrumentation validity concerns whether measures adequately and reliably represent the constructs they are intended to capture. This issue is particularly important in HRI because many frequently examined outcomes, including trust, anthropomorphism, perceived safety, acceptance, and related psychological constructs, are latent and cannot be directly observed.

The framework considers four features of subjective measurement. First, multi-item measures can provide more reliable representations of latent constructs than single-item measures and enable the empirical assessment of measurement reliability. Second, existing or adapted measures can draw on prior theoretical development and psychometric evidence associated with established instruments. Their use does not itself guarantee validity in a new

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from [permissions@acm.org](mailto:permissions@acm.org).

HRI '27, Santa Clara, CA, USA

© 2027 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-XXXX-X/2018/06

<https://doi.org/XXXXXXX.XXXXXXX>

population or context, but provides a basis on which measurement can build.

Third, reliability evidence provides information regarding the consistency of measurement. Depending on the measurement design, this may include internal consistency [7], test–retest reliability [11], inter-rater reliability, or other appropriate evidence. Fourth, construct validity concerns whether an instrument captures the theoretical construct it is intended to represent. Evidence may include convergent validity, discriminant validity, factorial validity, or other analyses explicitly used to establish construct validity [6, 20].

Because these criteria apply primarily to subjective constructs represented through measurement instruments, the instrumentation-validity indicators were evaluated among papers containing at least one subjective dependent variable.

## 1.4 Manipulation Validity

Manipulation validity concerns whether an experimental manipulation produces the intended distinction between experimental conditions. Evidence regarding the manipulation can help establish that the independent variable was instantiated as intended before differences in downstream outcomes are interpreted [4, 20].

The appropriate form of evidence depends on the nature of the manipulation. For objectively defined manipulations, a factual check may establish whether participants noticed or correctly identified the manipulated distinction. For manipulations intended to alter a subjective perception or psychological state, a perceptual measure of the manipulated construct may provide relevant evidence [12].

At the same time, the methodological role of manipulation checks depends on the experimental design and causal claim. Manipulation checks can differ in timing, reactivity, and theoretical function, and a measure of a proposed mechanism is not necessarily equivalent to a simple treatment-fidelity check. We therefore adopted a deliberately broad and descriptive coding criterion rather than judging whether a particular form of manipulation validation was necessary for each experiment. A paper was coded as reporting a manipulation check when it explicitly reported either a factual or perceptual check of an experimental manipulation.

## 1.5 External Validity

External validity concerns the extent to which experimental findings may generalize beyond the particular participants and contexts examined in a study. In HRI, potential boundaries on generalizability may arise from both the participant population and characteristics of the experimental context [14].

Rather than assigning each study a global judgment of external validity, we characterize observable features relevant to understanding the scope of the evidence. These include study setting, the modality through which participants encountered the robot, and the reporting of participant demographic and geographic information. We also examine whether authors explicitly discuss sample-related limitations to the generalizability of their findings.

This approach recognizes that no sample or experimental setting is inherently generalizable. The relevance of a particular population or context depends on the population and settings to which researchers intend their conclusions to apply. The coded variables

therefore characterize information that allows readers to understand the empirical context and evaluate potential boundaries of generalization rather than assigning studies a binary judgment of external validity.

## 1.6 Statistical Conclusion Validity

Statistical conclusion validity concerns whether the quantitative evidence provides an adequate basis for drawing conclusions about relationships among the variables examined. Statistical conclusions depend not only on whether an analysis produces a statistically significant result, but also on whether the sample and analytical procedures are appropriate for the intended inference [10, 18].

We focus on three methodological practices. First, sample-size justification may include an a priori power analysis, a sensitivity analysis, or another explicit rationale for selecting the sample size, such as evidence from a pilot study or reference to prior similar studies. These forms of justification provide different information about sample-size decisions, and the coding records their presence without treating any single form as proof of analytical adequacy. Second, statistical procedures rely on assumptions or model-quality criteria that differ across analytical approaches. Reporting relevant diagnostic procedures provides evidence that the analytical method was evaluated for appropriateness to the data and model. Third, effect sizes provide information about the magnitude of observed relationships or differences beyond statistical significance alone.

The framework does not prescribe a single statistical procedure or universal set of assumptions. Instead, coding focuses on whether papers report evidence appropriate to the analyses they employ.

## 1.7 Research Transparency

Research transparency concerns whether key methodological decisions and procedures are documented in ways that allow readers to understand and evaluate how an experiment was conducted. Transparent documentation also facilitates replication and cumulative scientific evaluation.

We examine two indicators in this dimension. Preregistration provides prospective documentation of research plans and analytical decisions and can help distinguish decisions specified before data analysis from those made subsequently. Institutional ethics or IRB approval provides documented evidence of institutional oversight of research involving human participants. Our coding concerns whether these practices were reported in the published paper or associated materials rather than making claims about whether an unreported practice was performed.

## 2 Codebook and Operational Criteria

### 2.1 General Coding Rules

Coding was conducted at the paper level. A methodological indicator was coded as present if the corresponding practice was explicitly reported for at least one applicable instance within a paper. For example, a paper was coded as reporting reliability if reliability evidence was provided for at least one applicable subjective measure. Similarly, a paper containing multiple manipulations was coded as reporting a manipulation check if at least one manipulation was explicitly checked. Coding was based on explicit evidence available in the published paper and, when available, its associated

supplementary materials. Coders evaluated each indicator independently according to the predefined coding criteria and based on whether the specified methodological practice was explicitly reported. Coders did not infer that a practice had been conducted when it was not reported. The coding criteria were designed to capture observable features of published reports rather than subjective judgments of methodological quality. For most methodological indicators, papers were coded as *Yes* when the specified evidence was present and *No* when it was absent. For study characteristics, coders extracted the reported methodological characteristics and recorded the corresponding category or information in the coding sheet.

## 2.2 General Paper Characteristics

### 2.2.1 Publication Venue.

*Definition.* The publication venue of the included paper.

*Code HRI Conference if.* Papers published in the ACM/IEEE International Conference on Human-Robot Interaction.

*Code THRI if.* Papers published in the ACM Transactions on Human-Robot Interaction.

### 2.2.2 Technical vs. Non-Technical Paper.

*Definition.* Whether the paper proposed a new system, design or algorithm in addition to conducting an experiment, or focused solely on the experiment.

*Code Technical if.* A new system, design, or algorithm was proposed and the subsequent experiment primarily evaluated that technical contribution. Thus, design-, technical-, or systems-oriented papers meeting this criterion were coded Technical.

*Code Non-technical if.* The paper did not introduce such a technical contribution and instead focused primarily on the user study.

### 2.2.3 Sample Size.

*Definition.* The final number of participants included in the analysis for each eligible experiment.

*Note:* The final sample size after exclusions was recorded rather than the initial recruitment count. When a paper contained multiple eligible experiments with distinct participant samples, participant counts were summed to obtain the paper-level total.

## 2.3 Theoretical Grounding

### 2.3.1 Explicit Hypothesis.

*Definition.* Whether the paper explicitly states at least one hypothesis regarding an expected relationship or difference.

*Code Yes if.* The paper explicitly states at least one hypothesis. Typically, the paper labels a prediction as a hypothesis (e.g., H1, H2) or explicitly identifies a statement as a hypothesis even if the hypotheses are not numbered.

*Code No if.* The paper does not explicitly state any hypotheses.

### 2.3.2 Theoretical or Logical Rationale for Hypotheses.

*Applicability.* This indicator was coded only for papers containing at least one explicit hypothesis.

*Definition.* Whether the paper provides an explicit theoretical or logical explanation for why at least one hypothesized relationship is expected to occur.

*Code Yes if.*

- At least one hypothesis is supported by an established theoretical framework.
- At least one hypothesis is supported by an explicit logical argument explaining why the proposed relationship should occur.

*Code No if.* Hypotheses are stated without an explanation for why the proposed relationship is expected.

*Note:* The coding criterion concerns the presence of theoretical or logical reasoning rather than the quality of that reasoning. Coders should examine both the text accompanying the hypotheses and relevant earlier sections, including Related Work, Background, and Literature Review sections. The rationale did not need to appear immediately alongside a hypothesis, but its connection to the predicted relationship had to be explicit.

### 2.3.3 Hypothesis Stated Before the Results Section.

*Applicability.* This indicator was coded only for papers containing at least one explicit hypothesis.

*Definition.* Whether at least one explicit hypothesis is stated before the paper's Results section or equivalent presentation of empirical findings.

*Code Yes if.* The hypothesis appears before the Results section.

*Code No if.* The hypothesis first appears in the Results, Discussion, or another section following presentation of the empirical findings.

*Note:* This indicator concerns the location of the hypothesis in the published report and does not establish when the hypothesis was originally formulated by the researchers.

## 2.4 Internal Validity

### 2.4.1 Random Assignment or Order Control.

*Definition.* Whether the paper reports an appropriate randomization or order-control procedure for its experimental design.

*Between-subject designs.* Code Yes if participants are explicitly reported as being randomly assigned or randomly allocated to experimental conditions.

*Within-subject designs.* Code Yes if the order of experimental conditions is explicitly reported as randomized, counterbalanced, or otherwise systematically varied to control order effects.

*Mixed designs.* Code Yes if the relevant between-subject assignment and/or within-subject order-control procedure is explicitly reported.

*Code No if.*

- Assignment or condition order is not described as randomized or controlled.

- The paper merely reports that participants were “assigned” to conditions without indicating random assignment.
- A within-subject study presents conditions in a fixed order without reported counterbalancing or randomization.

## 2.5 Instrumentation Validity

### 2.5.1 Presence of a Subjective Dependent Variable.

*Definition.* Whether the paper includes at least one dependent variable based on participants’ subjective self-report, perception, attitude, judgment, or evaluation.

*Code Yes if.* At least one outcome is measured using a subjective participant response, such as a questionnaire, rating scale, or self-reported evaluation.

*Code No if.* All dependent variables are objective or behavioral measures, such as task completion time, error rate, physical movement, or system performance, without a subjective dependent measure.

### 2.5.2 Multi-Item Measure.

*Applicability.* Papers containing at least one subjective dependent variable.

*Definition.* Whether at least one subjective construct is assessed using two or more items intended to represent the same construct.

*Code Yes if.* At least one subjective dependent construct is measured using multiple scale items.

*Code No if.* All subjective dependent constructs are assessed using single-item measures. For example, one question answered on a seven-point Likert scale is a single-item measure.

### 2.5.3 Existing or Adapted Measure.

*Applicability.* Papers containing at least one subjective dependent variable.

*Definition.* Whether at least one subjective dependent measure is explicitly described as existing, previously established, validated, or adapted from prior research.

*Code Yes if.*

- The paper uses a previously published scale or instrument.
- The paper explicitly states that a measure was adapted or modified from a previously published instrument.

*Code No if.*

- All subjective measures appear to be self-proposed without reference to an existing instrument.
- No information is provided regarding the origin of the measures.

### 2.5.4 Reliability Evidence.

*Applicability.* Papers containing at least one subjective dependent variable.

*Definition.* Whether the paper reports quantitative evidence explicitly used to establish the reliability or consistency of at least one subjective measure.

*Code Yes if.* The paper reports at least one recognized form of reliability evidence, including, for example:

- internal consistency, such as Cronbach’s  $\alpha$ ;
- composite reliability;
- test–retest reliability;
- split-half reliability;
- inter-rater reliability when applicable; or
- another quantitative statistic explicitly used as evidence of measurement reliability.

*Code No if.*

- No reliability evidence is reported for any subjective dependent measure.
- The paper merely cites prior reliability evidence for an instrument without reporting reliability evidence for its use in the present study.

### 2.5.5 Construct Validity Evidence.

*Applicability.* Papers containing at least one subjective dependent variable.

*Definition.* Whether the paper reports empirical evidence explicitly used to establish the construct validity of at least one subjective measure.

*Code Yes if.* The paper reports evidence such as:

- convergent validity;
- discriminant validity;
- factorial or structural validity;
- nomological validity; or
- another empirical analysis explicitly presented as evidence that the measure captures the intended construct.

*Code No if.*

- No construct validity evidence is reported.
- The paper uses an established instrument but provides no construct validity evidence for the measure in the present study.

*Note:* Validity examples include exploratory or confirmatory factor analysis explicitly used to evaluate the measurement structure, average variance extracted (AVE) used to assess convergent validity, and multitrait–multimethod (MTMM) analysis. For instance, a paper reporting factor loadings to assess the intended structure of a subjective scale or AVE-based analyses of convergent or discriminant validity would be coded Yes.

## 2.6 Manipulation Validity

### 2.6.1 Manipulation Check.

*Definition.* Whether the paper reports at least one explicit check of an experimental manipulation.

*Code Yes if.* At least one manipulation is accompanied by:

- a factual/attention check assessing whether participants recognized or identified the manipulated distinction; or
- a perceptual check assessing whether experimental conditions differed on the intended perceived construct.

*Code No if.*

- No manipulation check is reported.
- The paper only describes how the manipulation was implemented.
- The paper reports an attention check unrelated to whether the intended experimental manipulation was instantiated.

## 2.7 External Validity

External validity was characterized using both categorical descriptors of the experimental context and indicators describing the reporting of participant characteristics and generalizability limitations.

### 2.7.1 Study Setting.

*Definition.* The primary environment in which participants completed the experiment.

*Coding categories.*

- **Laboratory:** The experiment was conducted in a controlled laboratory or research facility.
- **Field:** The experiment was conducted in a naturally occurring or applied environment outside a conventional research laboratory.
- **Online:** Participants completed the experiment remotely through an online platform or web-based study. Examples include studies completed online through Prolific, CloudResearch, or Amazon Mechanical Turk.
- **Multiple:** The paper contained eligible experiments conducted in more than one of the above settings.
- **Not reported:** The paper did not report any information on the study setting.

### 2.7.2 Robot Interaction Modality.

*Definition.* The primary modality through which participants encountered or interacted with the robot during the experiment.

*Coding categories.*

- **Physical robot:** Participants directly encountered or interacted with a physically embodied robot.
- **Video/image/text-based:** Participants evaluated or interacted with a representation of a robot presented through video, images, written descriptions, or related non-immersive media rather than direct physical interaction.
- **AR/VR:** Participants encountered a robot through augmented-reality or virtual-reality technology.
- **Multiple:** The paper contained eligible experiments or study components involving more than one interaction modality and could not be represented by a single category.
- **Not reported:** The paper did not report any information on the interaction modality.

2.7.3 *Gender.* When the paper provided counts, percentages, or other sufficiently specific information describing participant gender, the numbers or percentages of female, male, and other reported gender categories were extracted for descriptive analysis.

2.7.4 *Age.* When the paper reported the mean age of the participant sample, the reported mean age was extracted for descriptive analysis. When sufficient information was available, the mean age was calculated from the reported sample information.

2.7.5 *Participant Country.* The participant country was coded as reported when the paper explicitly identified the country or countries from which participants were recruited or in which the study was conducted. The reported country information was then extracted. If no participant country was reported, the paper was coded as not reporting this information. Participant country was not inferred solely from author affiliations.

### 2.7.6 Sample-Related Limitation Reported.

*Definition.* Whether the paper discusses a limitation concerning the generalizability of its findings arising from characteristics of the participant sample.

*Code Yes if.* The authors identify a characteristic of their participant sample as limiting or potentially limiting the generalizability or external validity of the findings.

*Code No if.*

- No sample-related limitation to generalizability is discussed.
- The paper discusses general limitations unrelated to the participant sample.

## 2.8 Statistical Conclusion Validity

### 2.8.1 Sample-Size Justification.

*Definition.* Whether the paper reports at least one explicit justification for the selected sample size.

*Code Yes if.* At least one eligible experiment is accompanied by:

- an a priori power analysis or sample-size calculation;
- a sensitivity analysis evaluating the effect sizes detectable with the achieved sample; or
- another explicit rationale for the selected sample size, such as evidence from a pilot study or reference to prior similar studies.

*Code No if.* The paper reports only the achieved sample size without a power analysis, sensitivity analysis, or another explicit rationale for selecting that sample size.

### 2.8.2 Primary Statistical Analysis Approach.

*Definition.* The primary statistical approach used in the paper.

*Coding categories.*

- **Classical parametric tests:** For example, *t*-tests, ANOVA, ANCOVA, linear regression, and linear mixed models.
- **Non-parametric tests:** For example, Mann-Whitney *U*, Wilcoxon signed-rank, Kruskal-Wallis, Friedman, and Spearman rank correlation. The original scheme also grouped chi-square tests in this category.
- **SEM:** Structural equation modeling.
- **Other:** A primary approach not captured by these categories.

This field describes the analysis approach.

### 2.8.3 Statistical Assumption or Model-Quality Check.

**Definition.** Whether the paper reports checking at least one statistical assumption or model-quality criterion relevant to an analysis used in the study.

**Code Yes if.** The paper explicitly reports at least one applicable diagnostic or criterion.

- **Independent-samples *t*-test or ANOVA:** Independence, normality of model errors, and homogeneity of variance when required by the chosen procedure.
- **Repeated-measures ANOVA:** Relevant normality assumptions and sphericity when applicable.
- **ANCOVA:** Independence, residual normality, homogeneity of variance, linearity of the covariate–outcome relationship, and homogeneity of regression slopes for a common-slope model.
- **Linear or multiple regression:** Linearity, independence of errors, residual normality when relevant to inference, homoscedasticity, and absence of problematic multicollinearity where applicable.
- **SEM:** Measurement-model reliability and validity, as well as relevant model-fit or model-quality criteria.

**Code No if.** No statistical assumption check or model-quality criterion is reported.

#### 2.8.4 Effect-Size Reporting.

**Definition.** Whether the paper reports at least one quantitative measure of effect magnitude.

**Code Yes if.** The paper reports an effect-size statistic, such as Cohen’s  $d$ ,  $\eta^2$ , partial  $\eta^2$ ,  $R^2$ , a standardized regression coefficient ( $\beta$ ), Cramér’s  $V$ , an odds ratio, or a correlation coefficient ( $r$ ). Following the original coding notes, an explicitly reported standardized coefficient qualifies without requiring the authors to label it an effect size.

**Code No if.** The paper reports only  $p$ -values, test statistics (e.g.,  $t$ ,  $F$ , or  $\chi^2$ ), or raw means and standard deviations, without a separate effect-size statistic.

For example,  $t(58) = 3.42$ ,  $p = .001$ ,  $d = 0.89$  qualifies as Yes because Cohen’s  $d$  is reported. In contrast,  $F(1, 108) = 3.93$ ,  $p < .05$  without an accompanying effect size does not qualify.

## 2.9 Research Transparency

### 2.9.1 Preregistration.

**Definition.** Whether the paper reports that the study, hypotheses, design, or analysis plan was preregistered before the relevant empirical analysis.

**Code Yes if.** The paper explicitly states that the study or analysis was preregistered or provides a link or reference to a preregistration.

**Code No if.** No preregistration is reported.

### 2.9.2 IRB or Ethics Approval.

**Definition.** Whether the paper explicitly reports institutional ethics review or approval for the human-subject research.

**Code Yes if.** The paper states that the research was reviewed or approved by an Institutional Review Board, ethics committee, or equivalent institutional body.

**Code No if.** No institutional ethics review or approval is reported.

## 3 Additional Details for Table Categories

For study setting, five papers involved multiple settings and one did not report the study setting. Specifically, [2] included both online and laboratory experiments; [19] included an online experiment and another experiment for which the setting was not reported; [16] included both online and laboratory experiments; [8] included both online and laboratory experiments; and [1] included both laboratory and field experiments. [13] did not report the study setting.

For robot interaction modality, four papers could not be assigned to a single category and were therefore excluded from the single-category breakdown reported in the main manuscript. [3] combined AR/VR and physical robot interaction; [17] combined video/description-based and physical robot interaction; [19] combined a video/description-based experiment with another experiment whose robot-exposure modality was unclear; and [9] combined an image-based experiment with an audio-guided physical prototyping task.

## References

- [1] Rohan Banerjee, Krishna Palempalli, Bohan Yang, Jiaying Fang, Alif Abdullah, Tom Silver, Sarah Dean, and Tapomayukh Bhattacharjee. 2026. A Human-in-the-Loop Confidence-Aware Failure Recovery Framework for Modular Robot Policies. In *Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction* (Edinburgh, Scotland, UK) (HRI '26). Association for Computing Machinery, New York, NY, USA, 346–355. doi:10.1145/3757279.3788668
- [2] Adna Blik, M. Ege Cansev, and Philipp Beckerle. 2026. Human or Agent: Whom to Trust? Investigating Trust in Human–Robot Interaction Using Human Data to Train Reinforcement Agents. *J. Hum.-Robot Interact.* 15, 3, Article 68 (May 2026), 34 pages. doi:10.1145/3802818
- [3] Adna Blik, M. Ege Cansev, and Philipp Beckerle. 2026. Human or Agent: Whom to Trust? Investigating Trust in Human–Robot Interaction Using Human Data to Train Reinforcement Agents. *J. Hum.-Robot Interact.* 15, 3, Article 68 (May 2026), 34 pages. doi:10.1145/3802818
- [4] Marie-Claude Boudreau, David Gefen, and Detmar W Straub. 2001. Validation in Information Systems research: A state-of-the-art assessment1. *MIS quarterly* 25, 1 (2001), 1–16.
- [5] Donald T Campbell. 1957. Factors relevant to the validity of experiments in social settings. *Psychological bulletin* 54, 4 (1957), 297.
- [6] Donald T Campbell and Donald W Fiske. 1959. Convergent and discriminant validation by the multitrait-multimethod matrix. *Psychological bulletin* 56, 2 (1959), 81.
- [7] Lee J Cronbach. 1951. Coefficient alpha and the internal structure of tests. *psychometrika* 16, 3 (1951), 297–334.
- [8] Martina De Cet, Negin Hashmati, Mohammad Obaid, and Ilaria Torre. 2026. From Voice to Form: How Gender-Ambiguous Voices Shape Physical Robot Design. In *Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction* (Edinburgh, Scotland, UK) (HRI '26). Association for Computing Machinery, New York, NY, USA, 658–667. doi:10.1145/3757279.3785559
- [9] Martina De Cet, Negin Hashmati, Mohammad Obaid, and Ilaria Torre. 2026. From voice to form: How gender-ambiguous voices shape physical robot design. In *Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction*. 658–667.
- [10] Alan R Dennis and Joseph S Valacich. 2001. Conducting experimental research in information systems. *Communications of the association for information systems* 7, 1 (2001), 5.
- [11] William J Doll, Weidong Xia, and Gholamreza Torkzadeh. 1994. A confirmatory factor analysis of the end-user computing satisfaction instrument. *MIS quarterly* 18, 4 (1994), 453–461.
- [12] Klaus Fiedler, Linda McCaughey, and Johannes Prager. 2021. Quo vadis, methodology? The key role of manipulation checks for validity control and quality of science. *Perspectives on Psychological Science* 16, 4 (2021), 816–826.

- [13] Till Hielscher, Fabio Scaparro, and Kai O. Arras. 2026. Context-Aware Generation and Modulation of Expressive Motion Behavior using Multimodal Foundation Models. In *Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction* (Edinburgh, Scotland, UK) (HRI '26). Association for Computing Machinery, New York, NY, USA, 438–446. doi:10.1145/3757279.3785635
- [14] Guy Hoffman and Xuan Zhao. 2020. A primer for conducting experiments in human-robot interaction. *ACM Transactions on Human-Robot Interaction (THRI)* 10, 1 (2020), 1–31.
- [15] Norbert L Kerr. 1998. HARKing: Hypothesizing after the results are known. *Personality and social psychology review* 2, 3 (1998), 196–217.
- [16] Shipeng Lyu, Fangyuan Wang, Weiwei Lin, Guodong Guo, and David Navarro-Alarcon. 2026. Enhancing End-user Engagement in Human-Robot Interaction by Performing LLM-driven Expressive Behaviors. *J. Hum.-Robot Interact.* 15, 4, Article 79 (June 2026), 32 pages. doi:10.1145/3813107
- [17] Shipeng Lyu, Fangyuan Wang, Weiwei Lin, Guodong Guo, and David Navarro-Alarcon. 2026. Enhancing End-user Engagement in Human-Robot Interaction by Performing LLM-driven Expressive Behaviors. *J. Hum.-Robot Interact.* 15, 4, Article 79 (June 2026), 32 pages. doi:10.1145/3813107
- [18] Katherine Picho and Anthony R Artino Jr. 2016. 7 deadly sins in educational research. *Journal of Graduate Medical Education* 8, 4 (2016), 483–487.
- [19] Ian C. Rankin, Thane Somers, Sean Buchmeier, Alivia M. Eng, Cristina G. Wilson, and Geoffrey A. Hollinger. 2026. Explainable and Adaptable Robotic Decision Making for Scientific Data Collection. *J. Hum.-Robot Interact.* 15, 4, Article 75 (June 2026), 38 pages. doi:10.1145/3811028
- [20] Detmar Straub, Marie-Claude Boudreau, and David Gefen. 2004. Validation guidelines for IS positivist research. *Communications of the Association for Information systems* 13, 1 (2004), 24.
- [21] Detmar W Straub. 1989. Validating instruments in MIS Research1. *MIS quarterly* 13, 2 (1989), 147–169.